

AOC-1 orbital capsule

Specifications and assembly instructions



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Description

The AOC-1 is a low tech capsule that can be used for short suborbital or orbital missions. The capsule is powered by a small panel and has a redundant short range radio transceiver. Endurance is limited to 3 days.

Specifications

Parameter	Value	Unit	Notes
Mass	1161	Kg	Fully loaded
Total EC	310	Units	
EC consumption	0.11	EC/sec	Radios+computer+sensors
	0.095	EC/sec	2 radios and computer
	0.075	EC/sec	1 Radio and computer
	0.056	EC/sec	1 Radio, computer hibernating
	0.055	EC/sec	Computer only
	0.036	EC/sec	Life support only, all hibernating

Assembly and maintenance instructions

Parts list

1x	Mk1 Command Pod
1x	Probodobodyne OKTO
1x	OX-STAT Photovoltaic panel
1x	Heat Shield (1.25m)
1x	Mk16 Parachute
2x	Communotron 16-S
1x	2HOT Thermometer
1x	PressMat Barometer

Technology needed

The capsule needs a level 5 technology level on a single branch.

- Start
 - Command pod
 - Parachute
- Engineering 101
 - Thermometer
 - Radio
- Survivability

- Heat shield
- Barometer
- Electrics
 - Solar panel
 - OKTO

Assembly procedure

- Start with Mk1 pod
 - Drain all monopropellant
 - Remove all equipment from inventory slots
 - Set spare parts to 5
- Install heat shield axially on bottom of capsule
 - Set ablator to 0
- Install OKTO computer axially on top of capsule
 - OKTO name tag on window side
 - Set "Hibernate on warp" to "Auto"
 - Set "Hibernate" to "Off"
 - Set battery flow priority to -1
- Install radio antennas radially on pod
 - 2x Mirror symmetry
 - Move outward so that radio bodies lie at the edge of the sides of the OKTO panel
 - Rotate so that casing rests solidly on computer (at 0 degrees)
 - Move downwards until bottom of casings touch bottom of computer panel
 - Radio bodies should be centred in height
 - Antennas should embed into capsule and extend past computer
 - Antennas should be parallel to each other
- Install parachutes axially on top of OKTO
 - Angle 0
 - Altitude 1250
 - Pressure 0.6
- Install barometer and thermometer below parachutes
 - On top of white stripe on sides of capsule

- Barometer to left of hatch, thermometer to right
- Top of instrument touching eighth notch from the top
- Second notch after the upper panel break line
- Install OX-STAT panel
 - Line up mounting brackets with OKTO labeled panel
 - Same side as hatch

After assembly, consider keeping a copy of all items rooted in the OKTO and saving them as the capsule's computer section. This can be used for conversion of AOC-0 capsules into AOC-1.



Figure 1: AOC-1 showing position of instruments and radios

AOC-0 conversion instructions

To convert an AOC-0 capsule to the AOC-1 standard, perform the following procedures.

Before starting, have an AOC-1 computer section ready.

- Remove Stayputnik computer
- Remove parachutes
- Remove radios
- Reposition thermometer and barometer
 - Slide both up until the top of the instrument touches the eighth notch
- Install computer unit axially on top of Mk1 pod

General post flight maintenance

Part	System	Action	Notes
OKTO	Battery	Fill	

Part	System	Action	Notes
	Command	Check hibernation	Hibernate to off Hibernate in warp to on
Mk1 pod	Wheel	Active/Normal	
	Battery	Fill	
	Monoprop	Empty	
	Spare parts	5	
	Food	Fill	
	Water	Fill	
	Oxygen	Fill	
	Carbon Dioxide	Empty	
	Waste	Empty	
	Waste water	Empty	
Parachutes	Reaction wheels	Active/Normal	
	Pressure	0.6	
	Altitude	1250	
Radios	Deploy	When safe	
	Status	Enabled	

General pre mission checks

- Check parachute parameters
 - Pressure 0.6 Altitude 1250
- Check radios
 - Both ON
- Check supplies
 - EC full
 - Life support full
- Check computer
 - Not hibernating
 - Auto hibernate in warp

Systems

Electrical

Description

The electrical system is supplied by the Mk1 pod batteries. There is a secondary battery in the Stayputnik computer that is set to be drained after the main battery is drained. The batteries can be charged using the single solar panel present, endurance is not normally limited by electric charge, but by consumables.

System consumption

Class	System	Consumption	Unit	Notes
Computer	Total	0.02	EC/sec	
	OKTO	0.02	EC/sec	Full operation, divide by 1000 for hibernation
Radios	Total	0.04	EC/sec	
	C16-S (right)	0.02	EC/sec	
	C16-S (left)	0.02	EC/sec	
Life support Attitude	Total	0.04	EC/sec	
	Total	0.23	EC/sec	
	Mk1 wheel	0.2	EC/sec	
	OKTO	0.03	EC/sec	
Sensors	Total	0.0134	EC/sec	
	Thermometer	0.0067	EC/sec	Display active
		60	EC	Science transmission
	Barometer	0.0067	EC/sec	Display active
		90	EC	Science transmission

Total EC consumption

Situation	Consumption	Unit	Notes
Maneuvering	0.33	EC/sec	Sensors inactive
Full op + sensors	0.1134	EC/sec	
Full operation	0.10	EC/sec	
Hibernating	0.08	EC/sec	
Hibernating, 1 radio	0.06	EC/sec	
Hibernating, no comms	0.04	EC/sec	Requires crew to get out of this state

Life support

The life support is a standard TAC LS unit taking supplies from the Mk1 capsule. This is enough to keep a kerbonaut alive for 3 days. The capsule stores consumables for 3 days.

Resource	Endurance	Unit	Notes
Oxygen	18	hours	
Water	18	hours	
Food	18	hours	
Electric power	2.15	hours	Recharges using solar panel

Attitude control

Reaction Wheels

There is a redundant wheel system in the capsule. The Mk1 pod houses the primary wheel. There is a second wheel in the OKTO computer which can be used as an emergency control.

Orbital maneuvering

The AOC-1 is not capable of maneuvering in orbit and requires an external service module for all operations, including deorbiting.

Communications

The AOC-1 contains dual Communotron 16-S radios. These are enough to maintain communications when in view of the KSC. The system is capable of keeping in contact with the KSC at a distance of 7620 Km and with an HG-5 equipped satellite at 4479 Km.

Antenna ranges

(assumes root model, 0.5 antenna multiplier and 0.5 multiple antenna bonus)

Situation	Range	Unit	Power	Unit	Notes
Single radio	750	Km	0.02	EC/sec	
Dual radios	1125	Km	0.04	EC/sec	

To compute effective range, employ the root model equation.

$$R = \min(r_1, r_2) + \sqrt{r_1 \times r_2}$$

where r1 should be the capsule's range and the r2 range for an HG-5 is 10000Km and KSC is 37500Km.

Science

The capsule has a basic science instrument set containing a barometer and thermometer. These instruments have a storage capacity for one measurement. Data can be moved to the Mk1 pod via EVA or transmitted to KSC via radio.

Transmitting science data imposes a large power drain and should not be done without previous planning.

Re-entry

AOC-1 capsules can re-enter Kerbin's atmosphere passively. Once contact with the atmosphere is made, the capsule should orient itself in the proper shield first attitude. The shield does not use ablator.

Attaining re-entry attitude requires an external means of propulsion. The AOC-1 capsule is not capable of performing orbital maneuvers.

Landing

Landing is provided by a single Mk16 parachute. In case of failure, use the parachutes attached to the capsule service module, if available.

The main parachute allow a landing speed of 6.4 m/s at sea level. The parachute open partially at an altitude of 3000 m (0.6 pressure) and fully at 1250 metres. They need to be adjusted if the predicted course will cause the landing to happen on mountaineous terrain.

WARNING: Avoid mountains. Plan ahead before executing the re-entry procedure. Consider prematurely opening the parachutes to shorten downrange if necessary.